

Homework 4

EE 999 — Introduction to Control Systems

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Due Wednesday, October 7, 2026

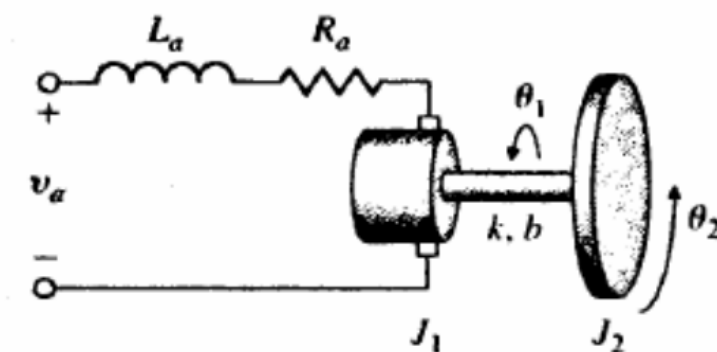
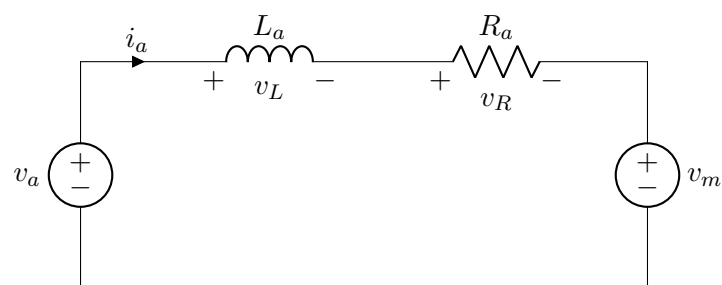
Problem A.1 — Equations of Motion
Figure 1: DC motor driving a load inertia J_2 

Figure 2: Electrical Circuit Model

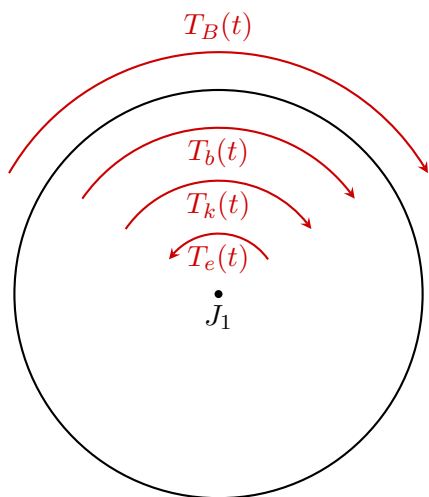


Figure 3: Free Body Diagram of Rotor

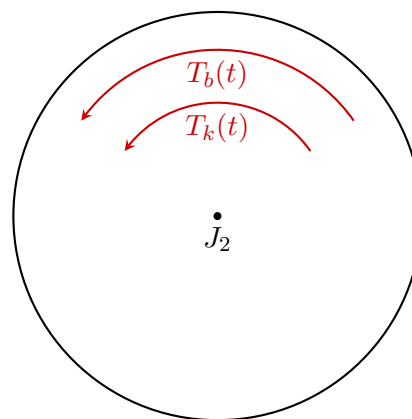


Figure 4: Free Body Diagram of Load

Electrical side:

$$v_a(t) = v_L(t) + v_R(t) + v_m(t)$$

$$v_L(t) = L_a \dot{i}_a(t)$$

$$v_R(t) = R_a i_a(t)$$

$$v_m(t) = K_e \dot{\theta}_1(t)$$

Rotor:

$$J_1 \ddot{\theta}_1(t) = T_{net.J_1}(t) = T_e(t) - T_B(t) - T_k(t) - T_b(t)$$

$$T_e(t) = K_t i_a(t)$$

$$T_B(t) = B \dot{\theta}_1(t)$$

$$T_k(t) = k(\theta_1 - \theta_2)$$

$$T_b(t) = b(\dot{\theta}_1 - \dot{\theta}_2)$$

Load:

$$J_2 \ddot{\theta}_2(t) = T_{net.J_2}(t) = T_b(t) + T_k(t)$$

Standard Forms

$$v_a(t) = L_a \dot{i}_a(t) + R_a i_a(t) + K_e \dot{\theta}_1(t)$$

$$J_1 \ddot{\theta}_1(t) = K_t i_a(t) - B \dot{\theta}_1(t) - k(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2)$$

$$J_1 \ddot{\theta}_1(t) + B \dot{\theta}_1(t) + b \dot{\theta}_1 + k \theta_1(t) = K_t i_a(t) + k \theta_2 + b \dot{\theta}_2$$

$$J_2 \ddot{\theta}_2(t) = k(\theta_1(t) - \theta_2(t)) + b(\dot{\theta}_1(t) - \dot{\theta}_2(t))$$

$$J_2 \ddot{\theta}_2(t) + b \dot{\theta}_2(t) + k \theta_2(t) = k \theta_1(t) + b \dot{\theta}_1(t)$$

Problem A.2 — Electrical Circuit Model

Mechanical	Units	Electrical	Units
$\dot{\theta}(t) = v(t)$	rad/s	Voltage	V
$T(t) = i(t)$	N·m	Current	A
$J = C$	kg·m ²	Capacitance	F
$B = 1/R$	N·m·s/rad	Conductance	S
$k = 1/L$	N·m/rad	Inverse inductance	1/H

Table 1: Firestone analogy: signal and unit correspondence.

Mechanical	Electrical
$T = J\ddot{\theta}$	$i = C\dot{v}$
$T = B\dot{\theta}$	$i = v/R$
$T = k\theta = k \int \dot{\theta} dt$	$i = \frac{1}{L} \int v dt$

Table 2: Element laws under the Firestone analogy.

Current equation for the first inertia:

$$\begin{aligned}
 J_1 \ddot{\theta}_1(t) &= K_t i_a(t) - B \dot{\theta}_1(t) - k(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2) \\
 k_t i_a(t) &= J_1 \ddot{\theta}_1(t) + B \dot{\theta}_1(t) + k(\theta_1 - \theta_2) + b(\dot{\theta}_1 - \dot{\theta}_2) \\
 &= C_1 \dot{v}_1(t) + \frac{1}{R_B} v_1(t) + \frac{1}{L_k} \left(\int v_1(t) - \int v_2(t) \right) + \frac{1}{R_b} (v_1(t) - v_2(t))
 \end{aligned}$$

and for the second:

$$\begin{aligned}
 J_2 \ddot{\theta}_2(t) &= k(\theta_1(t) - \theta_2(t)) + b(\dot{\theta}_1(t) - \dot{\theta}_2(t)) \\
 J_2 \ddot{\theta}_2(t) &= \frac{1}{L_k} \left(\int v_1(t) - \int v_2(t) \right) + \frac{1}{R_b} (v_1(t) - v_2(t))
 \end{aligned}$$

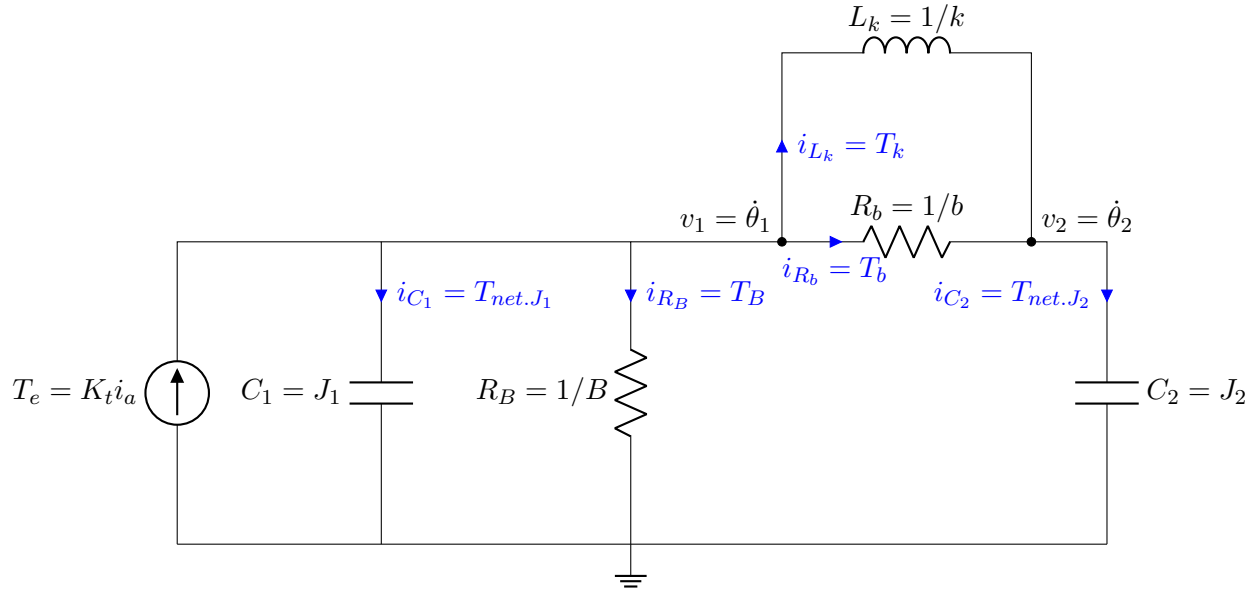


Figure 5: Firestone analogous circuit, with branch currents

Problem B — Units Balancing

$$J_m \ddot{\theta}_m + b \dot{\theta}_m = K_t i_a$$

$$(kg \cdot m^2) \left(\frac{rad}{s^2} \right) + \left(\frac{N \cdot m \cdot s}{rad} \right) \left(\frac{rad}{s} \right) = \left(\frac{N \cdot m}{A} \right) (A)$$

$$(N \cdot m \cdot \cancel{s}) \left(\frac{rad}{\cancel{s^2}} \right) + \left(\frac{N \cdot m \cdot \cancel{s}}{rad} \right) \left(\frac{rad}{\cancel{s}} \right) = \left(\frac{N \cdot m}{A} \right) (\cancel{A}) = N \cdot m$$

Noting that K_t is the motor torque constant with units of $\frac{N \cdot m}{A}$, so the amps cancel, and radians are eliminated as being unitless.